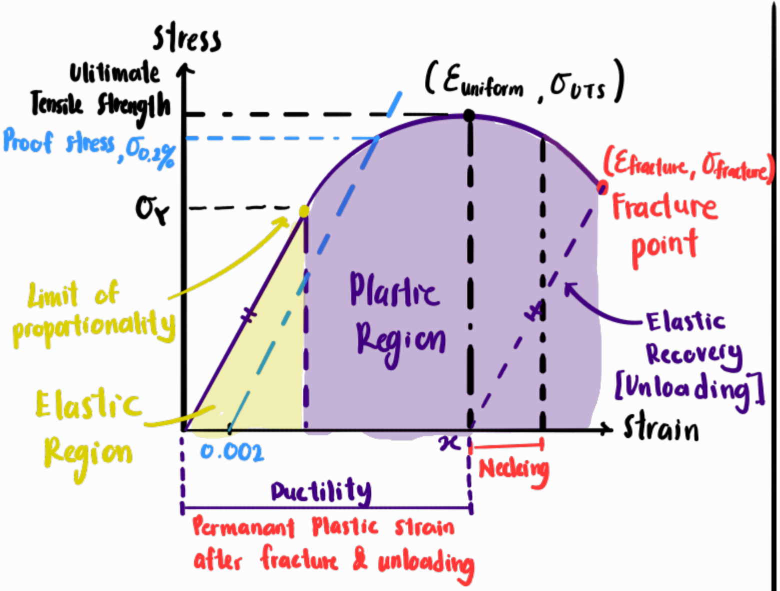


$$\sigma = \frac{F}{A}$$

$$\varepsilon = \frac{\Delta L}{L}$$

$$E = \frac{\sigma}{\varepsilon} = \frac{FL}{A \Delta L}$$

Thermoplastic	VS	Thermoset
· Ductility ↑		Ductility ↓
· Reversible		Irreversible

Toughness (Jm^{-3}) = $\frac{\text{Work Done}}{\text{Volume}}$

= Area under stress strain

E = Gradient of Elastic Recovery

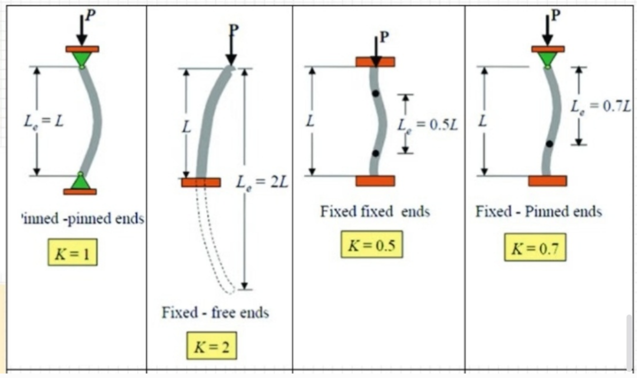
$= \frac{\sigma_{fracture} - 0}{\varepsilon_{fracture} - x}$

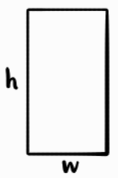
$$\text{Ductility}, x = \varepsilon_{fracture} - \frac{\sigma_{fracture}}{E}$$

Strain, $\varepsilon = \alpha(\Delta T)$

$$\% CW = \frac{A_0 - A_{final}}{A_0} \times 100$$

Buckling capacity, $P_{capacity} = \frac{\pi^2 E I}{(L_e)^2}$





$$I_{xx} = \frac{wh^3}{12}$$

$$I_{yy} = \frac{hw^3}{12}$$

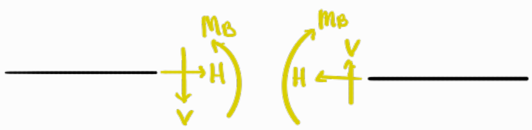
$$I = \frac{\pi r^4}{4}$$

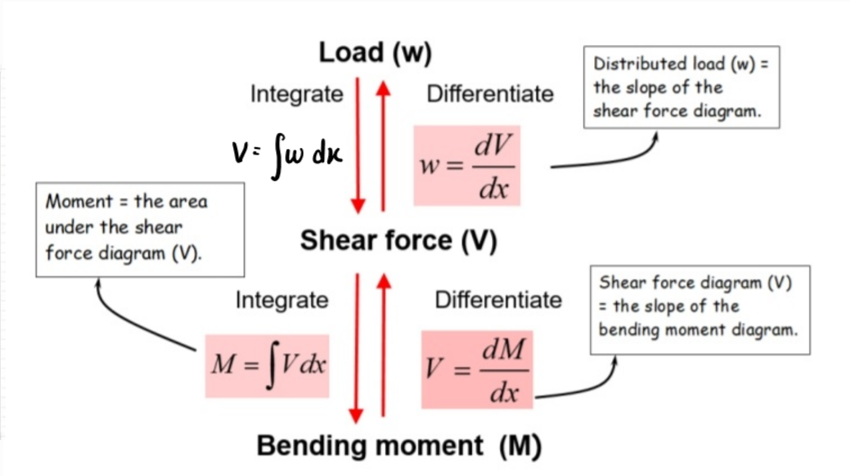
$$\sigma_{True} = \frac{F}{A_{actual}}$$

Cons. of volume : $A_0 L_0 = A_f L_f$

L_0 + fracture extension

Convention





1. The BM plots on the tension side of the beam.
2. M=0 at; a pin/roller end support, a cantilever end, a pt of inflexion and an internal pin.
3. Based on $V = dM/dx$;
   - '+V' = downwards slope in the BMD
   - '-V' = upwards slope in the BMD
   - Horizontal V --> BM is linear
   - Linear V --> BM is parabolic
   - V=0 --> BM is local max.or min
4. Based on $M = \int V dx$, moment = the area under SFD.

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\varepsilon_B = \frac{y}{R}$$

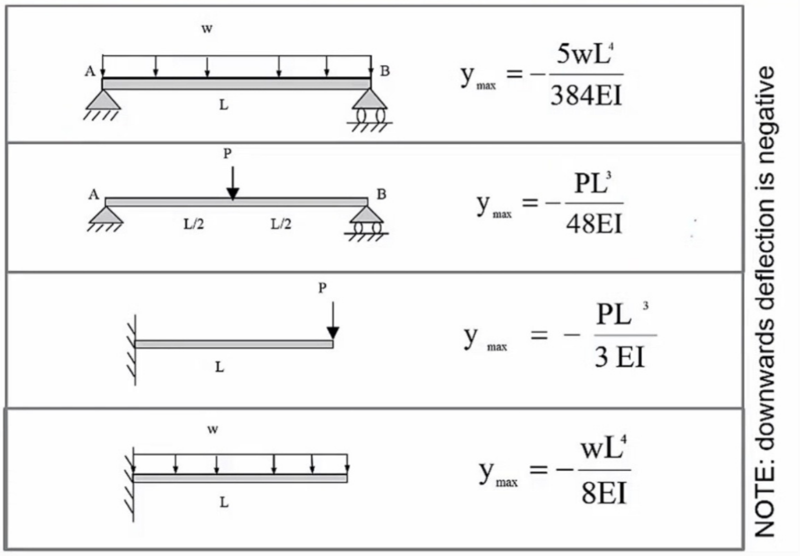
$$\sigma_B = \frac{Ey}{R} = \frac{M_B\, y}{I} = M/Z$$

$$M_B = \frac{EI}{R}$$

$$y_{neutral} = \frac{\Sigma A_i y_i}{A_{total}}$$

Dist btwn NA & centroid

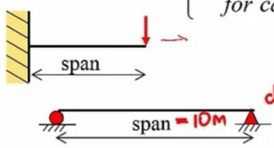
$$I_{new} = \Sigma I_{individual} + \Sigma A_i d_i^2$$



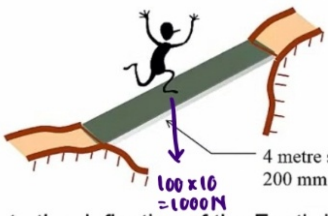
ULS ~ Load = 1.2 DL + 1.5 LL

SLS ~ Load = DL + 0.7 LL

Maximum Deflection < span/250 for beams; span/125 for cantilevers



defl < $\frac{10{,}000}{250}$ < 40mm



Loading = 100kg man at midspan.

Assume bridge to be weightless.

E_{timber} = 10,000MPa

Dead Load (DL) = 0

4 metre span footbridge.
200 mm x 40 mm timber plank.

Q. Calculate the deflection of the Footbridge at mid-span and check if the bridge meets the serviceability limit state requirements (deflect < span/250)

$$y_{max} = -\frac{PL^3}{48EI}$$

$$I = \frac{(200)(40)^3}{12} = 1.067 \times 10^6 \text{ mm}^4$$

With SLS, Load = DL + 0.7LL
= 0 + 0.7 (1000)
= 700N

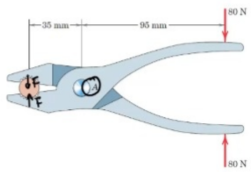
$$y_{max} = -\frac{(700\text{ N})(4000\text{mm})^3}{48(10{,}000\text{MPa})(1.067\times10^6\text{mm}^4)}$$

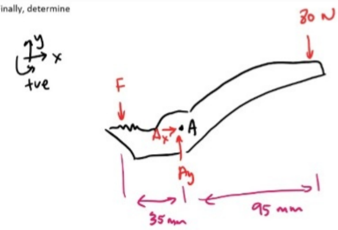
= 87.5 mm [Max deflection]

Allowable deflection = $\frac{\text{span}}{250} = \frac{4000}{250}$ = 16mm

∴ Since 87.5 > 16, the bridge fails SLS conditions.

applied on the rod by each of the plier jaws. Additionally, calculate the forces in the pin at point A. Finally, determine the mechanical advantage of this simple machine.





$\Sigma F_x = 0$

$A_x = 0$

$\Sigma F_y = 0$
$-F + A_y - 80 = 0$
$-F + A_y = 80$
$A_y = 80 + F$
$= 80 + 217.1429$
$A_y = 297.1429$ N

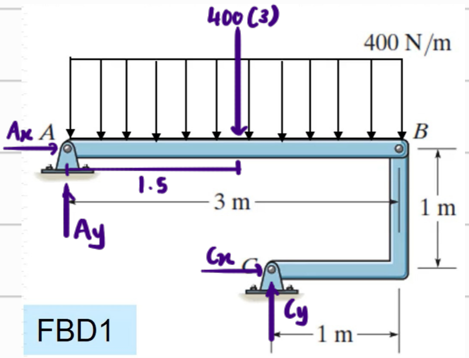
$\Sigma M_A = 0$
$F(35) - 80(95) = 0$
$F = 217.1429$ N

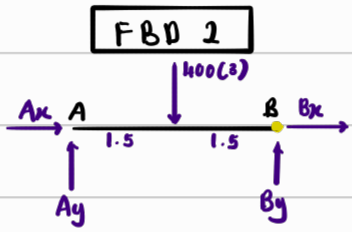
$M_A = \frac{F_{output}}{F_{input}} = \frac{217.1429}{80}$

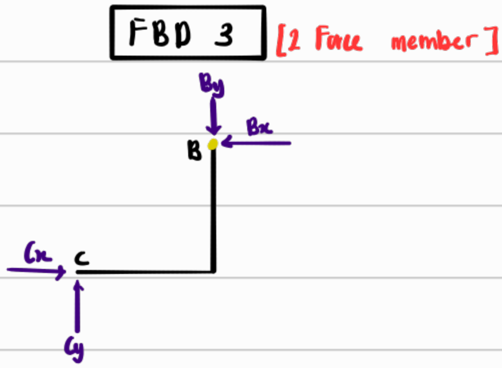
$M_A = 2.7143$

Example 1

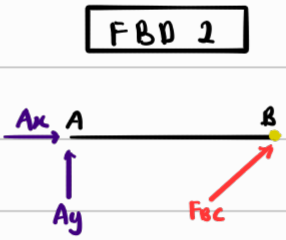
Solve for all unknown forces

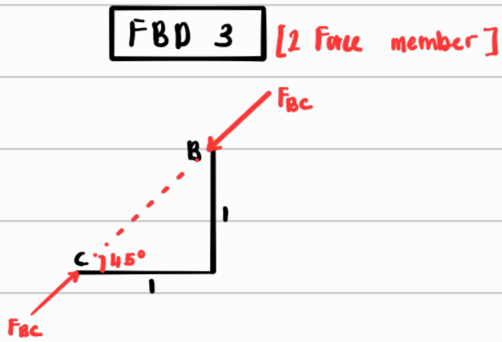




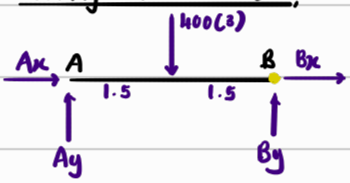


Since FBD 3 is a 2 force member, the FBDs can be redrawn,





① Looking at FBD 2,



By symmetry, $A_y = B_y$

$\therefore \; A_y + B_y - 400(3) = 0$

$2A_y = 1200$

$A_y = 600\text{ N}$

$\therefore \; B_y = 600\text{ N}$

② Looking at FBD 1

$\Sigma F_y = 600 - 1200 + C_y$

$C_y = 600\text{ N}$

$\Sigma M_A = -(1200 \times 1.5) + (600 \times 2) + C_x(1)$

$C_x = 600\text{N}$

$\Sigma F_x = A_x + 600$

$A_x = -600\text{ N}$

③ Looking back at FBD 2

$\Sigma F_x = A_x + B_x$

$0 = -600 + B_x$

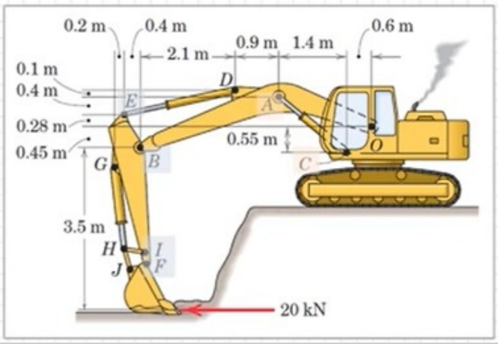
$B_x = 600\text{ N}$

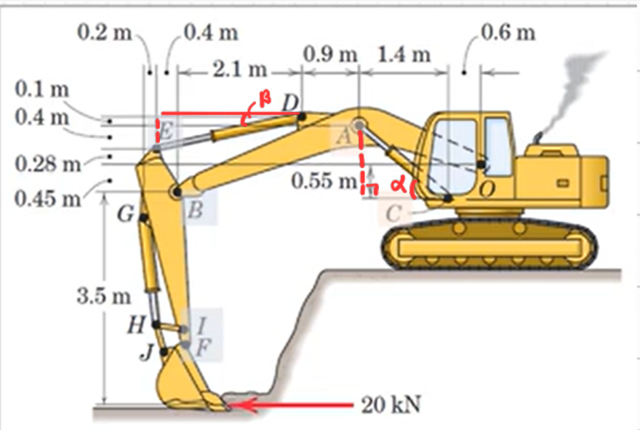
This method does not utillise the 2 force member, however u can if u wanted to

Example 2 :

Machine example

The excavator applies a 20 kN force parallel to the ground. There are two hydraulic cylinders at AC to control the OAB arm and a single cylinder DE to control the EBIF arm. Determine the force in the hydraulic cylinders AC. & DE





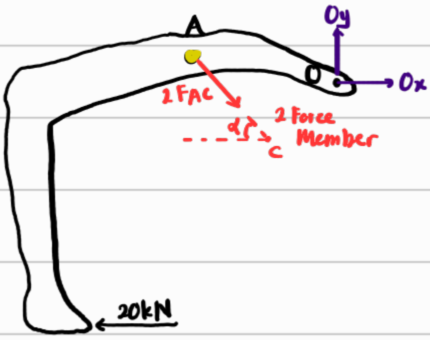
$$\tan\alpha = \frac{0.55 + 0.28 + 0.4}{1.4}$$

$$\alpha = 41.3°$$

$$\tan\beta = \frac{0.1 + 0.4}{0.4 + 2.1}$$

$$\beta = 11.31°$$

FBD 1



$$\Sigma M_O = 0$$

$$-20{,}000(3.95) - [2F_{AC}\cos(41.3°)][0.68] + [2F_{AC}\sin(41.3°)][2] = 0$$

$$\boxed{F_{AC} = 48.8\text{kN}}\ \#$$

FBD 2

$$\Sigma M_B = 0$$

$$-20{,}000(3.5) + [F_{DE}\cos(11.31°)](0.73) + [F_{DE}\sin(11.31°)](0.4) = 0$$

$$\boxed{F_{DE} = 88.1\text{kN}}\ \#$$

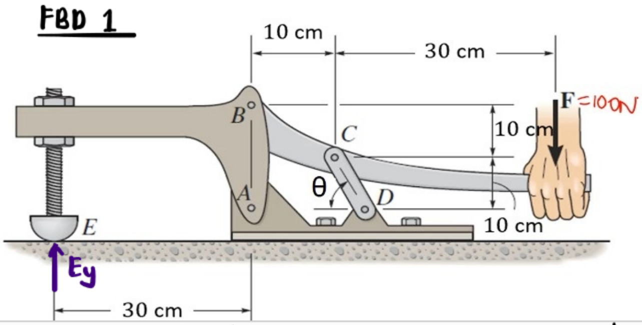
Example 3 :

Toggle clamp Example

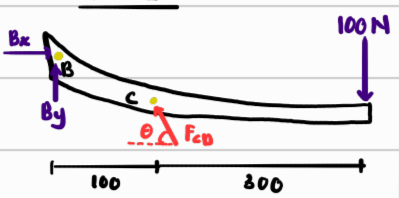
A toggle clamp is a type of mechanical clamp that is commonly used in manufacturing and woodworking applications to hold workpieces in place during machining or assembly processes.

A toggle clamp is subjected to a force $F = 100\ N$ at the handle. Determine the magnitude of the vertical clamping force acting at E and the mechanical advantage of this machine.

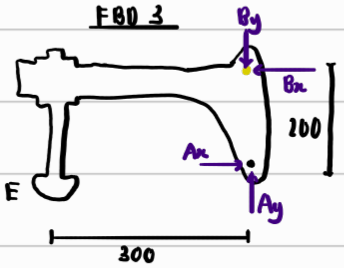
FBD 1



FBD 2



FBD 3



FBD 4

Looking at FBD 2

$\Sigma M_B = 0$

$-100(400) + (F_{CD}\sin\theta)(100) - (F_{CD}\cos\theta)(100) = 0$

$\boxed{F_{CD} = \dfrac{400}{(\sin\theta - \cos\theta)}}$ —①

$\Sigma F_x = 0$

$B_x - F_{CD}\cos\theta = 0$

$\boxed{B_x = F_{CD}\cos\theta}$ —②

$\Sigma F_y = 0$

$B_y + F_{CD}\sin\theta - 100 = 0$

$\boxed{B_y = 100 - F_{CD}\sin\theta}$ —③

Looking at FBD 3

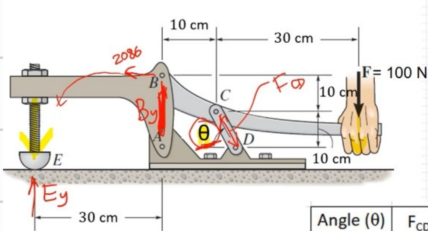
$\Sigma M_A = 0$

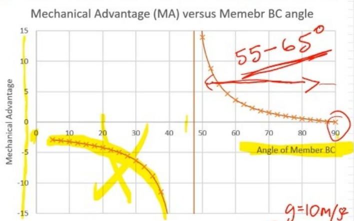
$(B_x)(200) - (E_y)(300) = 0$

$E_y = \dfrac{200 B_x}{300}$

$\boxed{E_y = \dfrac{800\cos\theta}{3(\sin\theta - \cos\theta)}}$ [using ① & ②]

Forces & Mechanical advantage for different angles





g=10m/s²
139 kg

θ= 50, 55, 60,

Angle (θ)	F_{CD} (N)	B_x (N)	B_y (N)	E_y (N)	MA
50	3245.3	2086.0	-1986.0	1390.7	13.9
55	1628.8	934.3	-834.3	622.8	6.2
60	1092.8	546.4	-446.4	364.3	3.6
65	827.0	349.5	-249.5	233.0	2.3

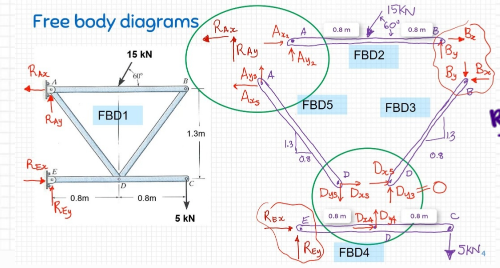
Example 4:

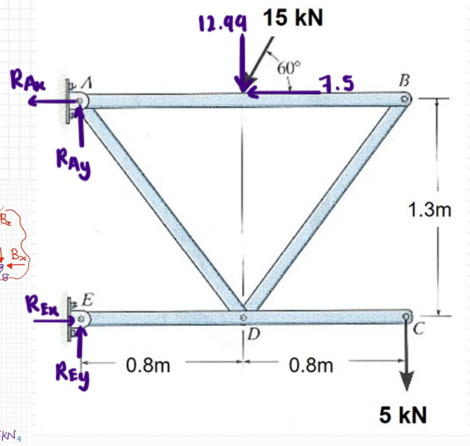
Four member frame

For the frame shown;

a) Find the support reactions

b) Calculate the axial forces in members AD and BD





① **Looking at FBD 1**

$\Sigma M_E = 0$

$1.3 R_{Ax} - (12.99)(0.8) + (7.5)(1.3) - (5)(1.6) = 0$

$R_{Ax} = 6.6476\,kN$ #

$\Sigma F_x = 0$

$R_{Ex} - 6.6476 - 7.5 = 0$

$R_{Ex} = 14.1475\,kN$ #

$\Sigma F_y = 0$

$R_{Ay} - 12.99 - 5 + R_{Ey} = 0$

$R_{Ay} + R_{Ey} = 17.99$ —— ①

② **Looking at FBD 4**

$\Sigma M_D = 0$

$-(5)(0.8) - R_{Ey}(0.8) = 0$

$R_{Ey} = -5\,kN$ #

∴ Using ①,

$R_{Ay} - 5 = 17.99$

$R_{Ay} = 22.99\,kN$ #

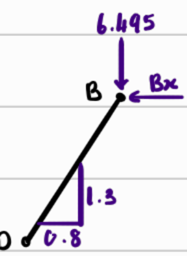
③ **Looking at FBD 2**

By symmetry, $A_{y_2} = B_y = \frac{12.99}{2} = 6.495\,kN$

$\Sigma F_x = 0$

$A_{x_2} - 7.5 + B_x = 0$ —— ②

④ **Looking at FBD 3**



Since this is a 2 force member, the forces must align with the axis of the member,

$\therefore \frac{1.3}{0.8} = \frac{6.495}{B_x}$

$B_x = 4\,kN$ #

$\therefore F_{BD} = \sqrt{(6.495)^2 + 4^2}$

$= 7.628\,kN$ (compression)

∴ Using ②,

$A_{x_2} - 7.5 + 4 = 0$

$A_{x_2} = 3.5\,kN$

Looking at FBD 5

$\Sigma F_x = 6.6476$

$3.5 + A_{x_5} = 6.6476$

$A_{x_5} = 10.147\,kN$

$\Sigma F_y = 22.99$

$6.495 + A_{y_5} = 22.99$

$A_{y_5} = 16.495\,kN$

$\therefore F_{AD} = \sqrt{(10.147)^2 + (16.495)^2}$

$= 19.366\,kN$ (Tension)

$$\boxed{\begin{aligned} \therefore F_{BD} &= 7.628\ (C) \\ F_{AD} &= 19.366\ (T) \end{aligned}}$$

Question 2

A sample of high-density polyethylene is tested in a tensile testing machine. When measured in an environment at 20 °C, the sample has an original cross-sectional area of 14.4 mm^2 and a gauge length of 25 mm.

The resulting force and extension readings can be downloaded from the following link: HDPE.xlsx

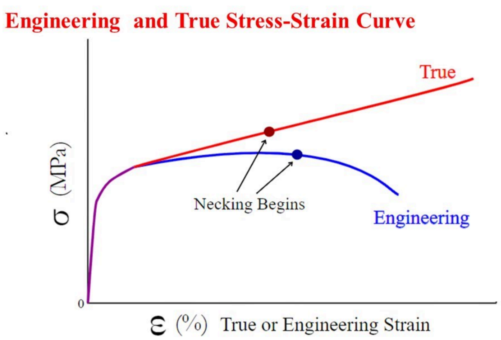
(a) Determine the ultimate tensile stress of the material a) σ_{max} from table = 19.3569 MPa

(b) Estimate the ductility of the material b) $E = \frac{0.7032}{0.0017} = 413.65 MPa \implies 413.65 = \frac{16.1521 - 0}{0.4947 - x} \implies x = 0.456 = 45.6\%$

(c) Estimate the toughness of the material c) Area under graph $= \frac{1}{2}(19.3569 \times 0.1607) + \frac{1}{2}(19.3569 + 16.1521)(0.4947 - 0.1607)$
$= 7.49 \, Jm^{-3}$

(d) Assuming the volume of the sample is constant, determine the true stress at the point where the sample breaks

d) $(14.4)(25) = A_f(25 + 12.368)$
$A_f = 9.6339 \, mm^2$
$\therefore \sigma_{true} = \frac{232.59}{9.6339} = 24.14 \, MPa$



$\sigma_{ENG} = \frac{F}{A_0}$ and $\sigma_{TRUE} = \frac{F}{A_i}$

At fracture:

$$\sigma_{TRUE} = \frac{F}{A_f}$$

To conserve volume,

$$A_0 L_0 = A_f L_f$$

where Lf = 25mm + fracture extension

e) Since it has a high ductility, HDPE is a thermoplastic polymer. Injection molding for high volume

(e) Is HDPE a thermoplastic polymer or a thermoset polymer? Name a manufacturing process, discussed this semester, that might be appropriate for producing a batch of 200,000 HDPE items.

(f) HDPE has a linear thermal expansion coefficient of 150 x 10^{-6}/°C. The HDPE part is exposed to an elevated temperature of 80 °C (from room temp: 20°C), which is still well below the melting point of HDPE. Will the sample be larger or smaller? Calculate the new cross-sectional area of the sample at this temperature.

$\varepsilon = \alpha \Delta T$
$= (150 \times 10^{-6})(60°C)$
$= 9 \times 10^{-3}$
$= 0.009$

$A = (1.009)^2 (A_0)$
$= 25.45 \, mm^2$ #
$\therefore$ Larger.

Question 3

Consider the pin-ended truss shown below with two vertical point loads P and Q. Based on this loading the greatest tension and compression forces in members BC and GF respectively are as follows;

$$F_{BC} = 95 \text{ kN (T)} \quad \text{and} \quad F_{GF} = 100 \text{ kN (C)}$$

$\sigma = \frac{F}{A}$

a) (i) Member **BC** is to be designed as a steel square hollow section. Based on applying a strength factor of safety (FOS) of 2.2, which minimum weight member in the Table below would you choose for this section? What will be the actual FOS of your chosen member? $F_{capacity} = 95 \times 2.2 = 209\text{kN}$, $350 = \frac{209}{A}$, $A = 597\text{mm}^2$ | Choose $65 \times 65 \times 2.5$SHS, $A = 609\text{mm}^2$; $SF_{actual} = \frac{350 \times 609}{95} = 2.24$ #

(ii) Based on your section choice in part a) what is the change in length of this member based on the unfactored axial load experienced? Provide your answer in units of mm. $E = \frac{FL}{A\Delta L} \Longrightarrow \Delta L = \frac{95k\,(4)}{(609 \times 10^{-6})(200{,}000M)} = 3.12\text{mm}$ #

b) (i) Member **GF** is to be a custom made steel circular hollow section of outer diameter 110mm. Based on applying a strength factor of safety of 2.2 again, what is the minimum required thickness for this section based on a strength limit state design. $P_{capacity} = 100 \times 2.2 = 220\text{kN}$, $350 \geq \frac{220}{A}$, $A \geq 628.52\text{mm}^2$ | $P \leq \frac{\pi^2 EI}{Le^2}$, $I = \frac{(220{,}000)(1 \times 4)^2}{\pi^2(200{,}000M)} \geq 1.78 \times 10^6\text{mm}^4$; $\frac{\pi(55)^4}{4} - \frac{\pi(r)^4}{4} \geq 1.78 \times 10^6$, $r \leq 51.22\text{mm}$ (Dominant) $\therefore$ Thickness $55 - 51.22 = 3.78\text{mm}$

$\pi(110^2) - \pi(r^2) \geq 628.52$

$r \leq 109\text{mm}$

(ii) Based on the section chosen in part (b)(i) above at what member length will the failure mode change from buckling to yielding? Draw the axial compression failure curve showing this value.

Assume the following values for the Material Properties: Young's modulus E = 200,000 MPa and Yield strength = 350MPa

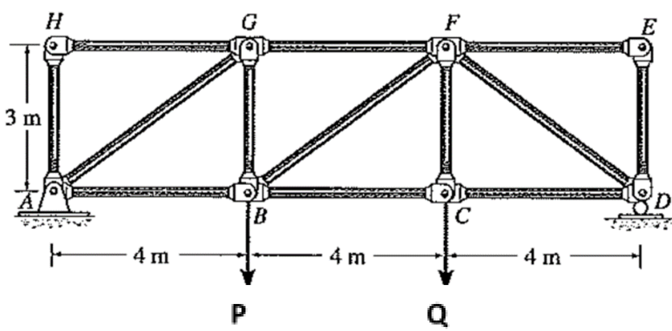
b) ii) Squash load $= (350)[\pi(55^2 - 51.22^2)]$

$= 441{,}485\text{N}$

$P_{capacity} = \frac{\pi^2(200{,}000M)(1.78 \times 10^6)}{(1 \times L)^2}$

$L^2 = \frac{\pi^2(200{,}000M)(1.78 \times 10^6)}{441{,}485}$

$L = 2.82\text{m}$



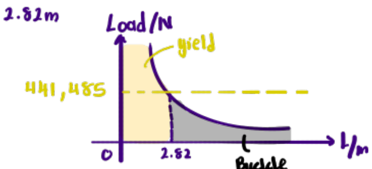


Table 1. Dimensions and properties of square hollow sections of dimensions b x b x t (AS/NZS 1163:2016)

Designation: Depth b (mm)	Width b (mm)	Thickness t (mm)	Mass per unit length m (kg/m)	External surface area: Per unit length A_{EL} (m^2/m)	External surface area: Per unit mass A_{EM} (m^2/t)	Ratio $\frac{b-2t}{t}$	Gross area of cross-section A_g (mm^2)	Second moment of area I_x, I_y (10^6 mm^4)
100 ×	100 ×	9.0 SHS	23.5	0.361	15.4	9.11	3000	3.91
100 ×	100 ×	6.0 SHS	16.7	0.374	22.4	14.7	2130	3.04
100 ×	100 ×	5.0 SHS	14.2	0.379	26.6	18.0	1810	2.66
100 ×	100 ×	4.0 SHS	11.6	0.383	32.9	23.0	1480	2.23
100 ×	100 ×	3.0 SHS	8.96	0.390	43.5	31.3	1140	1.77
89 ×	89 ×	6.0 SHS	14.6	0.330	22.5	12.8	1870	2.06
89 ×	89 ×	5.0 SHS	12.5	0.334	26.7	15.8	1590	1.81
89 ×	89 ×	3.5 SHS	9.06	0.341	37.6	23.4	1150	1.37
75 ×	75 ×	6.0 SHS	12.0	0.274	22.8	10.5	1530	1.16
75 ×	75 ×	5.0 SHS	10.3	0.279	27.0	13.0	1310	1.03
75 ×	75 ×	4.0 SHS	8.49	0.283	33.3	16.8	1080	0.882
75 ×	75 ×	3.5 SHS	7.53	0.285	37.9	19.4	959	0.797
75 ×	75 ×	3.0 SHS	6.60	0.290	43.9	23.0	841	0.716
75 ×	75 ×	2.5 SHS	5.56	0.291	52.4	28.0	709	0.614
65 ×	65 ×	3.0 SHS	5.66	0.250	44.1	19.7	721	0.454
65 ×	65 ×	2.5 SHS	4.78	0.251	52.6	24.0	609	0.391
65 ×	65 ×	2.0 SHS	3.88	0.253	65.3	30.5	494	0.323
50 ×	50 ×	4.0 SHS	5.35	0.183	34.2	10.5	681	0.229
50 ×	50 ×	3.0 SHS	4.25	0.190	44.7	14.7	541	0.195
50 ×	50 ×	2.5 SHS	3.60	0.191	53.1	18.0	459	0.169
50 ×	50 ×	2.0 SHS	2.93	0.193	65.8	23.0	374	0.141
50 ×	50 ×	1.6 SHS	2.38	0.195	81.7	29.3	303	0.117

Question 4

A cantilever beam has a uniformly distributed load, a point load and an applied moment acting on it. The point load and moment both act at location B in the beam. The resulting support reactions at A are;

$A_x = 0$ kN, $A_y = 15$ kN (upwards) $A_m = 44.25$ kNm (anticlockwise)

a) Draw a free body diagram (FBD) of the left section of the beam, with the cut made just to the <u>left</u> of point B (i.e. 1.99999m from A). This FBD will thus exclude the point load and moment acting at B. Using your FBD and applying the equilibrium equations $\Sigma F_y = 0$ and $\Sigma M_z = 0$, calculate the value of the shear and the moment values in the beam at this location.

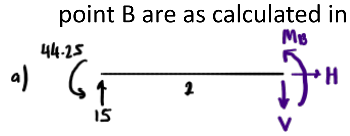
b) Draw a free body diagram (FBD) of the left section of the beam, with the cut made just to the <u>right</u> of point B (i.e. 2.00001m from A). This FBD will now include the point load and moment acting at B. Using your FBD and applying equilibrium equations, calculate the value of the shear and the moment values in the beam at this location.

c) Draw a FBD of the left section of the beam, now cutting through the beam between points B and C. In your FBD define the distance between point A and the cut using the parameter 'x', where 2m < x < 4.5m and x = 0 lies at A. Using your FBD and applying equilibrium equations, derive the general mathematical expressions for the shear force and bending moment in the beam in terms of the parameter 'x'.

(For additional practice, in your own time, repeat a) and b) considering a FDB of the beam right of the cuts. Do the same for part c) and derive general equations where the origin of 'x' now lies at the cantilever tip C. To check equivalency of your general equations substitute in numerical values of 'x' i.e. x = 2.6m from A will be equivalent to x = 1.9 from point C)

d) Draw the Shear force and Bending moment diagrams and sketch the corresponding deflected shape. It is recommended you draw your SFD and BMD aligned with the beam based on the lines shown below. Check your values of shear and moment either side of point B are as calculated in part a) and b).

a)



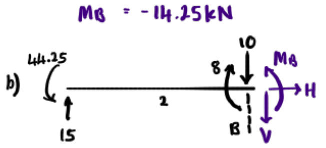
$\Sigma F_y = 15 - V$

$V = 15kN$

$\Sigma M_{cut} = 44.25 - 2(15) + M_B$

$M_B = -14.25kN$

b)



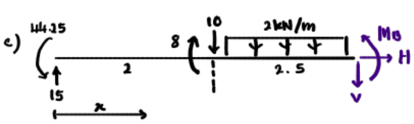
$\Sigma F_y = 15 - 10 - V$

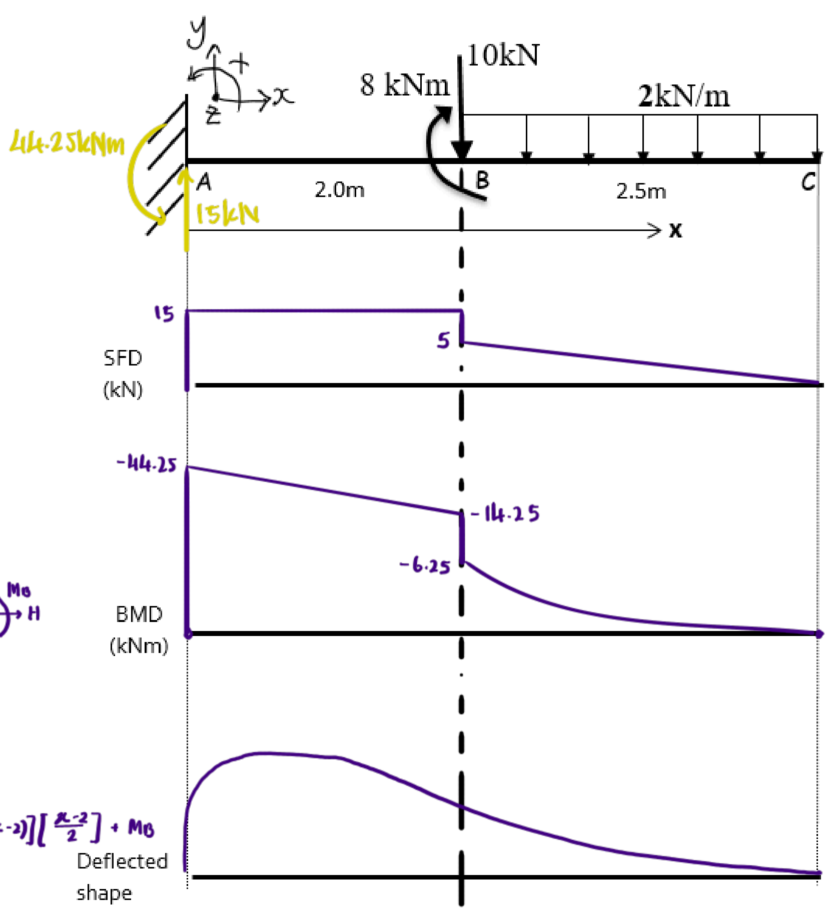
$V = 5kN$

$\Sigma M_{cut} = 44.25 - 2(15) - 8 + M_B$

$M_B = -6.25kNm$

c)





<u>$2 < x < 4.5$</u>

$\Sigma F_y = 15 - 10 - [(x-2)(2)] - V$

$V = 5 - 2x + 4$

$V = -2x + 9$

$\Sigma M_{cut} = 44.25 - 15x - 8 + 10(x-2) + [2(x-2)][\frac{x-2}{2}] + M_B$

$-M_B = 16.25 - 5x + x^2 - 4x + 4$

$= x^2 - 9x + 20.25$

$M_B = -x^2 + 9x - 20.25$

Question 5

For the beam shown below;

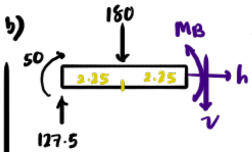
a) Calculate the support reactions.

a) $\Sigma M_A = -50 - 320(4) + 8B_y - 11(20) - 150$

$B_y = 212.5\,kN$ #

$\Sigma F_y = A_y + B_y - 320 - 20$

$A_y = 127.5\,kN$ #

b) 

$\Sigma M_c = M_B + 2.25(180) - 4.5(127.5) - 50$

$M_B = 218.75\,kNm$ #

$\Sigma F_y = 127.5 - 180 - v$

$v = 52.5\,N$ #

b) Draw a free body diagram (FBD) of the left section of the beam, with the cut made 4.5m to the right of support A. Using your FBD and applying the equilibrium equations $\Sigma F_y = 0$ and $\Sigma M_z = 0$, calculate the value of the shear and the moment at this location.

(For additional practice, in your own time, you can consider an FBD to the right of the cut including support B.)

c) Draw a FBD of the left section of the beam, now cutting through the beam between points A and B. In your FBD define the distance between point A and the cut using the parameter 'x', where 0m < x < 8.0m and x = 0 lies at A. Using your FBD and applying equilibrium equations, derive the general mathematical expressions for the shear force and bending moment in the beam in terms of the parameter 'x'. Check your answer in part b) by substituting x=4.5m into your general moment equation.

(For additional practice, in your own time, derive equations where the origin lies at the cantilever tip C. You can check the equivalency of your equations by substituting in numerical values.)

d) Draw the Shear force and Bending moment diagrams and sketch the corresponding deflected shape. It is recommended you draw your SFD and BMD aligned with the beam based on the lines shown below.

c) $\Sigma F_y = -v - 40x + 127.5$

$v = -40x + 127.5$

$\Sigma M_c = -50 + 40x\left(\frac{x}{2}\right) - 127.5x + M_B$

$M_B = -20x^2 + 127.5x + 50$

when $x = 4.5$

$M_B = 218.75$ ✓